SAHAYAK - PRESENTATION GUIDE

SHYAMANTIKA PEGU

INTRODUCTION, PROBLEM AND PURPOSE

Suggested speaking time: 3-4 minutes

Speaking script, slide cues, handover, troubleshooting and judge Q&A

YOUR ROLE

You are the first speaker. Set the context, explain the communication challenge, introduce SAHAYAK and its limited goal, then hand over clearly to Barakha. Keep the technical details for the second speaker.

SLIDE 1 - TITLE AND OPENING

Say: Good morning respected judges, teachers, and everyone present here. We are presenting SAHAYAK - a real-time Indian Sign Language recognition system. Our project explores how computer vision and machine learning can support communication accessibility. We will explain the problem, show how our prototype works, and discuss what it can and cannot do.

Stage cue: Pause after the project name and introduce your teammates.

SLIDE 2 - THE COMMUNICATION CHALLENGE

Say: People who use sign language can face barriers when people around them do not understand the signs they use. This can make everyday conversations more difficult in classrooms, public services, and other settings. Technology may help in particular situations, but it must be developed with the people who use it.

Important distinction: Indian Sign Language is a language, not just a collection of hand gestures. Sign languages have their own grammar and variation. Recognising a few hand signs is not the same as understanding a full signed conversation.

SLIDE 3 - WHY ACCESSIBILITY MATTERS

Optional evidence: The 2011 Census of India recorded 5,071,007 people in its 'disability in hearing' category. The WHO's hearing-loss fact sheet discusses the wider global impact of hearing loss. These figures describe hearing-related disability; they are NOT counts of Indian Sign Language users.

Say: These figures remind us that hearing accessibility is an important issue, but we must not confuse hearing-loss statistics with the number of people who use ISL. Our project addresses one small technical problem within a much larger accessibility challenge.

Sources: https://censusindia.gov.in/nada/index.php/catalog/43369/ and https://www.who.int/news-room/fact-sheets/detail/deafness-and-hearing-loss

SLIDE 4 - INTRODUCING SAHAYAK

Say: SAHAYAK is a camera-based prototype. It detects hand landmarks, turns them into numerical features, and sends those features to a trained machine-learning model. The model predicts a label from the signs it has learned. The website then displays the result.

Simple explanation: Landmarks are reference points on a hand. Instead of asking the model

to interpret a whole picture directly, the application describes hand position and shape using numbers. The model learns patterns in those numbers from labelled examples.

SLIDE 5 - PROJECT GOAL AND SCOPE

Say: Our goal is to demonstrate a working pipeline from camera input to a predicted sign label. The current system has a limited vocabulary. It is not a complete ISL translator, and it should not be treated as a replacement for interpreters or qualified communication support.

The website has supporting speech and translation functions, but not every feature works offline. The system may make mistakes because of lighting, camera angle, hands being hidden, movement, or differences between signers.

HANDOVER TO BARAKHA

Say: I have introduced the problem and why we chose this project. My teammate Barakha will now explain the technical workflow, how hand landmarks become features, and how the machine-learning model makes a prediction.

Stage cue: Turn toward Barakha and pause for the next speaker.

JUDGE QUESTIONS AND ANSWERS

Q: Is sign language universal? A: No. Different sign languages exist, including Indian Sign Language, and they have their own linguistic rules.

Q: Does SAHAYAK translate every ISL sentence? A: No. It recognises a limited set of trained labels; full language translation is a much harder problem.

Q: Can it replace an interpreter? A: No. It is an educational prototype, not a substitute for qualified interpreters.

Q: Why is the project useful if it has limitations? A: It demonstrates a possible assistive-technology approach and gives us a way to study recognition, evaluation, and responsible design.

PRESENTATION CHECKLIST

Practise the opening and handover aloud. Keep eye contact instead of reading every sentence. If you do not know a statistic, leave it out rather than guess. When discussing hearing-loss figures, clearly say they do not represent ISL users. If a judge asks for accuracy data that has not been measured, say so honestly.

ONE-SENTENCE SUMMARY

I explain why SAHAYAK was built, what problem it explores, and why the current prototype is a limited sign-recognition demonstration rather than a complete language translator.